Core practice | True / false

Scope: Lectures 1-9 in your Midterm folder, consistent with your existing revision guide. No separate official midterm notice was found. Original question numbers and wording are preserved. Choose a = True or b = False. Historical exam rules and marks are not current midterm instructions.

1. In C++, a **variable** is a named address for storing a type of value, and you must declare a variable before you can use it.

   a. True b. False

2. In C++, a **function** is a named section of code that performs a specific operation, and a function can include only one **`return`** statement as the exit point of the function.

   a. True b. False

3. The following two statements will generate two strings with the same content:

```
std::string str1 = "hello,world";
std::string str2 = "hello" + "," + std::string("world");
```

   a. True b. False

4. The insertion operator **<<** is often overloaded in different classes to insert the contents of the classes into an output stream for display purposes. The overloading usually requires the use of **return by reference**, because it avoids making a copy of the output stream. If you use **return by value**, a new copy of the output stream will be returned.

   a. True b. False

5. In a collection class, the **effective size** can never be equal to or greater than the **allocated size**.

   a. True b. False

6. The following statements declare two pointers **`p1`** and **`p3`**, and two integers **`p2`** and **`p4`**:

```
int*  p1, p2;
int  *p3, p4;
```

   a. True b. False

7. For a variable **`x`**, the expression **`*&x`** is essentially equivalent to **`x`**; and for a pointer **`p`**, the expression **`&*p`** is equivalent to **`p`**.

   a. True b. False

Core practice | Short questions

Q9: true / false. Q11, Q12, Q14 and Q15: choose ONE option. Q15 has been joined across its original page break.

9. If `c` is a nonempty collection in C++ STL, calling `c.begin()` and `c.end()` will return **iterators** pointing at the first and last element of that collection, respectively.

   a. True b. False

11. Calculate the result of the following expressions based on the C++ rules:

```
9 * (8-7)/6 + 5*4 % 3*(2+1.0)
```

   a. `3`
   b. `3.5`
   c. `7`
   d. `7.5`

12. In the definition of a class, it is preferred that the data members be declared as:

   a. public
   b. protected
   c. private
   d. friend

14. If we have built a **36**-bit computer on which the memory address will be represented in one 36-bit long **machine word**, and the **smallest addressable unit** is a sequence of 8 consecutive bits called a byte, how much memory can this computer have, theoretically? (Assume 1KB = 2^{10} bytes, 1MB = 2^{10} KB, 1GB = 2^{10} MB.)

   a. 36 GB
   b. 64 GB
   c. 256 GB
   d. 2^{36} GB

15. Assuming the following C++ variable declaration statement

```
int i = 2021, *pi = &i, **ppi = &pi;
```

   leads to the addresses of `i` and `pi` being `0x2020` and `0x2024`, respectively, what are the values of `ppi` and `**ppi`?

   a. `0x2020` and `2021`
   b. `0x2024` and `2021`
   c. `0x2020` and `0x2024`
   d. `0x2024` and `0x2028`

Core practice | Concepts

Q20: choose ONE option. Q21, Q22, Q24 and Q25: choose one or more options.

20. If the parameter you are passing is an object whose value should not change, and you don't want to make a copy of the object because it occupies a lot of memory, which one of the following approaches should you use to pass this object?

    a. Call by value
    b. Call by reference
    c. Call by pointer
    d. Constant call by reference

21. What are the **distinguishing** features of **object-oriented** programming languages like C++ (i.e., what makes object-oriented programming different from other programming paradigms)?

    a. Abstraction
    b. Encapsulation
    c. Inheritance
    d. Polymorphism

22. In C++, functions can be **overloaded**, which means that you can define different functions with the same name as long as they have different **signatures**. The signature of a function includes the following:

    a. The number of the parameters
    b. The names of the parameters
    c. The types of the parameters
    d. The type of the return value

24. Indicate which of the following are **illegal** variable names in C++:

    a. `typename`
    b. `type_name`
    c. `type-name`
    d. `2typenames`

25. Which of the following are the areas of memory in which values can be stored in a C++ program?

    a. Static area
    b. Heap
    c. Stack
    d. Queue

Core practice | Memory and classes

Q26, Q27 and Q30: choose one or more options. Q30 tests which function a base-class pointer calls.

26. Normally in C++ you should not use pointers as much as in the C language, in which pointers are important for the following reasons:

    a. Pointers make it possible to reserve new memory during program execution.
    b. Pointers allow you to specify the memory addresses for dynamically allocated variables and objects in the heap.
    c. Pointers allow you to refer to a large data structure in a compact way.
    d. Pointers can be used to record relationships among data items, e.g., linked list.

27. In C++, any expression that refers to an internal memory location capable of storing data is called an **lvalue**, which can appear on the left side of an assignment statement. Which of the following terms in a C++ program **cannot** be lvalues?

    a. `p`
    b. `*p`
    c. `&p`
    d. `(p+1)`

30. After running the following program, which of the numbers will be seen in the output?

```
#include <iostream>
using namespace std;
class A {
public:
    int a;
    A() { a = 1; }
    void display() { cout << a << endl; }
};
class B: public A {
public:
    int b;
    B() { b = 2; }
    void display() { cout << a << b << endl; }
};
class C: public B {
public:
    int c;
    C() { b = 1; c = 3; }
    virtual void display() { cout << a << b << c << endl; }
};
int main() {
    C oC;
    B* pb = &oC;
    pb->display();
    return 0;
}
```

    a. `1`
    b. `2`
    c. `3`
    d. None, compile error.

Core practice | Pointer tracing

Q31 has been joined across its original page break. Preserve the exact printed output, including line breaks. Use the remaining space for a trace table.

31. What is the formatted output of the following program? Write your answer in the box, and make sure the format is correct. (12%)

```
#include <iostream>
using namespace std;

int main(void)
{
    int arr[] = {2, 0, 2, 1};
    int* p = arr;
    int a = *++p;
    int b = *p++;
    int c = ++*p;
    int d = (*p)++;
    cout << arr[0] << arr[1] << arr[2] << arr[3] << endl;
    cout << a << b << c << d << endl;
    return 0;
}
```

Core practice | Array addresses

Q32: the first entry is already supplied in the original, leaving 12 blanks. Distinguish pointer addresses from numeric element values. Workings can go below.

32. Assume that **doubleArray** is declared as

```
double doubleArray[] = {0, 1, 1, 2, 3, 5, 8, 13, 21, 34};
```

and variables of type double take up **8 bytes** on the computer system you are using. If the base address of the array **doubleArray** happens to be **0xACE0**, what are the values of the following 12 expressions? Write your answer in the boxes below, **all in 4-character hexadecimal format**, e.g., 0 should be 0000. (12 x 1.5% = 18%)

`doubleArray`	= 0x	ACE0
`*doubleArray`	= 0x	
`&doubleArray`	= 0x	
`&doubleArray[0]`	= 0x	
`doubleArray[0]`	= 0x	
`doubleArray+1`	= 0x	
`&doubleArray+1`	= 0x	
`&doubleArray[1]`	= 0x	
`doubleArray+9`	= 0x	
`*doubleArray+9`	= 0x	
`*(doubleArray+9)`	= 0x	
`&doubleArray[9]`	= 0x	
`doubleArray[9]`	= 0x	

Optional | Confirm coverage first

These are related to the lectures but not firmly supported as core scope: Q10 (functional programming), Q17 (bit masks; operators appear in precedence tables), Q23 (Stanford-library-specific range-for support). Q10 is true/false; Q17 single choice; Q23 multiple choice.

10. C++ is a **multi-paradigm** programming language. Besides the **procedural** programming paradigm originally supported by C, C++ also supports the **object-oriented** paradigm. However, C++ does not support other paradigms such as the **functional** paradigm without external libraries.

    a. True b. False

17. Computers typically represent each pixel in an image as a 32-bit integer, in which the 4 bytes are interpreted as the α (Transparency), Red, Green, and Blue components. The following bitwise operations extracts one of the components from the hexadecimal representation of the αRGB color stored in an integer variable **`pixel`**. Which color is extracted and what is the component value in **decimal** format?

```
int pixel = 0xFF202104;
int color = (pixel >> 8) & 0xFF;
```

    a. α, 0
    b. α, 255
    c. Green, 21
    d. Green, 33

23. Which of the following collection classes in the Stanford C++ Library do not support the use of the range-based **`for`** loop?

    a. **`Vector`**
    b. **`Stack`**
    c. **`Queue`**
    d. **`Map`**